

Lava — Bug Fix Audit Trail

Per constitutional clause §6.T.4 (**Bugfix Documentation**) — every bug fix in this project **MUST** be documented here with root cause analysis, affected files, fix description, link to the verification test/ challenge, and the commit SHA that landed the fix.

§6.O (Crashlytics-Resolved Issue Coverage Mandate) extends this for Crashlytics-recorded issues; their closure logs live at `.lava-ci-evidence/crashlytics-resolved/<date>-<slug>.md`. §6.T.4 covers the rest — operator-reported, self-discovered, or reviewer-flagged bugs that don't enter the Crashlytics pipeline.

Format per entry:

`## YYYY-MM-DD - <short slug>`

`**Root cause:** ...`

`**Affected files:** ...`

`**Fix:** ...`

`**Verification test/challenge:** path or commit ref`

`**Fix commit:** SHA`

`**Forensic anchor:** (optional) what surfaced the bug`

2026-05-06 — phase1-distribute-three-bugs

The first real-distribute of Lava-Android-1.2.7-1027 + lava-api-go-2.1.0 exposed three bugs that landed in commit e947081:

Root cause 1: `/health` and `/ready` endpoints were registered AFTER the auth middleware in `lava-api-go/cmd/lava-api-go/main.go` `buildRouter`, so the orchestrator's liveness probe got 401 → restart loop.

Root cause 2: `core/network/impl/.../LavaAuthBlobProvider.kt` was declared internal, but the Phase 11 build-time-generated `lava.auth.LavaAuthGenerated` class lives in `:app`'s source set — different module → cannot access internal interface → compile failure.

Root cause 3: `scripts/distribute-api-remote.sh` + `deployment/thinker/thinker-up.sh` did NOT ship the operator's local `.env` `LAVA_AUTH_*` + `LAVA_API_HTTP3_*/BROTLI_*/PROTOCOL_*` values to the `thinker.local` container. The new 2.1.0 binary required `LAVA_AUTH_FIELD_NAME` + `LAVA_AUTH_HMAC_SECRET` at boot → crash-loop.

Affected files:

- `lava-api-go/cmd/lava-api-go/main.go`

- core/network/impl/src/main/kotlin/lava/network/impl/LavaAuthBlobProvider.kt
- scripts/distribute-api-remote.sh
- deployment/thinker/thinker-up.sh

Fix:

1. Register /health + /ready BEFORE the auth chain.
2. Drop internal from LavaAuthBlobProvider.
3. Distribute script merges operator's .env auth/transport block into a temp env file before scp; thinker-up iterates the variables and passes each as -e VAR=\$VAR to podman run.

Verification test/challenge: post-fix smoke test in scripts/distribute-api-remote.sh's 60-second /health wait (passes); curl -fsSk https://thinker.local:8443/{health,ready} return {"status":"alive"} / {"status":"ready"} (passes); auth gate returns 401 with {"error":"unauthorized"} for missing header (passes — fail-closed posture confirmed).

Fix commit: e947081

Forensic anchor: the very first bash scripts/distribute-api-remote.sh after the api-go-2.1.0 build showed lava-api-go: config: config: LAVA_AUTH_FIELD_NAME is required in a tight loop, with the orchestrator marking the container (unhealthy) and triggering the restart loop. Operator ran the distribute, got the lock-up output, immediately surfaced the issue.

2026-05-12 — c03-anonymous-toggle-checkauth-bluff

Root cause: OnboardingViewModel.onTestAndContinue() (the "Test & Continue" button handler on the Configure step) called sdk.checkAuth(currentId) for BOTH AuthType.NONE providers AND config.useAnonymous=true paths, then asserted that the result equals AuthState.Authenticated. For users opting INTO anonymous mode on a FORM_LOGIN tracker (RuTor with the toggle on), checkAuth correctly returns Unauthenticated — that IS the user's chosen state. The code treated it as a failure (error = "Connection failed") and never advanced to the Summary step. The wait-for-"All set!" timeout fired and the user was stuck.

The C03 Challenge Test detected this only as a 60s timeout — no indication of WHY. The bug was invisible to the Sixth Law clauses 1-5 because:

- The test's primary assertion was on "All set!" appearing (✓ correct per clause 3).
- The production stack was traversed end-to-end (✓ correct per clauses 1 + 4).
- But there was no logged stack trace; the catch block called analytics.recordNonFatal (Crashlytics-only) and silently set the error state. The diagnostic gap was the source of the bluff: green test infrastructure + green code review + green Bluff-Audit on prior commits all missed it because nobody had run C03 end-to-end on a device with logcat tailing to see the error.

Affected files:

- feature/onboarding/src/main/kotlin/lava/onboarding/OnboardingViewModel.kt

Fix:

1. Skip `sdk.checkAuth(currentId)` entirely on the anonymous branch. Anonymous = user opted out of auth = no auth state to check = not a failure when there's no session.
2. Added diagnostic `logger.d/logger.e` breadcrumbs on the entire `onTestAndContinue` flow (auth path taken, result, exception). This closes the §6.T.4-spirit gap that made the C03 bug invisible — future failures will print stack traces to logcat.
3. Added `HttpTimeout`, `UserAgent`, and `Logging` plugins to the `rutracker Ktor HttpClient` + 30s timeouts on the main `OkHttpClient`. The HTTP improvements did NOT close the C02 issue (Cloudflare-side stall — see §4.5.3b in `CONTINUATION.md`), but they raise the time-budget for slow networks and move the failure mode from `SocketTimeoutException` to a more diagnosable signal.
4. Bumped `Challenge02` test's wait for "All set!" from 30s to 90s to align with realistic real-network round-trip times after the timeout fixes.

Verification test/challenge:

- C03 (`Challenge03AnonymousSearchOnRuTorTest`) — PASS on CZ_API34_Phone API 34 (live emulator at localhost:5555), 2026-05-12 09:25:14 to 09:25:23, total ~9.7s. Evidence at `.lava-ci-evidence/Lava-Android-1.2.13-1033/post-mortem/logcat-c03-fix.txt` shows anon path: `switchTracker(rutor) → test ok: advance to next/Summary → Perform Finish`.
- Falsifiability rehearsal: re-introducing the broken `if (result != null && result != AuthState.Authenticated)` check in the anonymous branch causes C03 to time out at 60s with the Configure screen still visible (the pre-fix failure mode). Verified through the diagnostic logcat: `checkAuth(rutor) result=Unauthenticated → error path triggered → no advance`.

Fix commit: *(this commit)*

Forensic anchor: Phase 1 systematic-debugging session 2026-05-12. Operator picked option C ("investigate C02/C03 properly") + invoked the anti-bluff mandate. Matrix runs on CZ_API34_Phone surfaced the 60s timeout; live-emulator instrumentation captured the anon path: `checkAuth(rutor) result=Unauthenticated` line that identified the bug.

2026-05-12 — onboarding-action-logger-leaks-credentials

Root cause: Discovered during the C03 investigation above — `OnboardingViewModel.perform()` logged the action via `logger.d { "Perform $action" }`. For sealed-class subtypes `OnboardingAction.UsernameChanged(value: String)` and

OnboardingAction.PasswordChanged(value: String), Kotlin's auto-generated toString includes the value, so logcat (and any analytics breadcrumb that picks it up) prints the operator's real RuTracker username + password in plain text. Severity: §6.H credential inviolability concern — although .env is gitignored and the values never reach disk via a log file in the normal user flow, on a device with adb logcat running (or with logcat being collected by a 3rd party app, or shared in a bug report) this is a leak surface.

Affected files:

- feature/onboarding/src/main/kotlin/lava/onboarding/OnboardingViewModel.kt

Fix: changed the log expression from logger.d { "Perform \$action" } to logger.d { "Perform \${action::class.simpleName}" } — prints only the action type name, never any of its values.

Verification test/challenge: logcat tail during a re-run of C02 shows Perform UsernameChanged (no value) and Perform PasswordChanged (no value).

Fix commit: *(same commit as the c03-anonymous-toggle-checkauth-bluff entry above)*

Forensic anchor: logcat captures during the C02/C03 diag run on 2026-05-12 surfaced the bug. Pre-fix logcats are gitignored under .lava-ci-evidence/**/post-mortem/ to prevent accidental commit of the captured credentials; only matrix attestations + gradle.logs + JUnit XML test-reports are committed.

2026-05-12 — rutracker-cloudflare-mitigation + cookie-selection-bug

Root cause 1 (Cloudflare anti-bot stall): provideRuTrackerHttpClient in :core:tracker:client constructed a Ktor + OkHttp HttpClient with no cookies, no Accept-Language, no Accept, no Accept-Encoding, and no explicit User-Agent — i.e. a client whose HTTP/2 header shape is a flag-raising fingerprint to Cloudflare's anti-bot. A bare POST to rutracker.org/forum/login.php was accepted by the TCP+TLS layer but never received a response body — verified via Ktor Logging plugin output. Host-side curl with normal browser-like headers returns HTTP/2 200 in <1s for the same URL.

Root cause 2 (cookie selection picked wrong cookie as session token): RuTrackerInnerApiImpl.login() previously extracted token = cookies.firstOrNull { !it.contains("bb_ssl") } from the final-hop Set-Cookie headers. When Cloudflare in front of rutracker started emitting cf_clearance=... on every response, this filter started selecting cf_clearance as the "rutracker session token". The subsequent mainPage(token) GET then sent Cookie: cf_clearance=... without the actual bb_data session token, so rutracker returned the guest page and parseUserId couldn't find the logged-in element.

Affected files:

- core/tracker/client/src/main/kotlin/lava/tracker/client/di/TrackerClientModule.kt
- core/tracker/rutacker/src/main/kotlin/lava/tracker/rutacker/impl/RuTrackerInnerApiImpl.kt

Fix:

1. Installed HttpCookies plugin (AcceptAllCookiesStorage) on the rutacker HttpClient.
2. Added browser-class defaultRequest headers (Accept, Accept-Language, Accept-Encoding) + Chrome 124 / Android 14 / Pixel 8 User-Agent.
3. Added a runCatching { httpClient.get(Index).bodyAsText() } pre-flight inside RuTrackerInnerApiImpl.login() so the cf_clearance cookie lands in the cookie jar before the POST.
4. Tightened the post-login token extraction to match by name prefix (bb_data / bb_session / bb_login) instead of the fragile !contains("bb_ssl") negation.

Verification test/challenge: Ktor wire-log on C02 re-run shows

- 10:16:58 GET /forum/index.php → 200 (~1s) — pre-flight succeeds, cookies stored
- 10:16:59 POST /forum/login.php → 302 (was: 60s timeout)
- 10:16:59 GET /forum/index.php (auto-redirect) → 200 (~1.3s) — logged-in page returned

The Cloudflare stall is fully resolved by this commit. **C02 itself does NOT yet pass end-to-end** because GetCurrentProfileUseCase.parseUserId throws on the post-login page — #logged-in-username element isn't in the served HTML (selector stale or mobile-shaped variant). That's a separate domain-archaeology task documented in docs/CONTINUATION.md §4.5.3c.

Fix commit: (*this commit*)

Forensic anchor: continued Phase 1 systematic-debugging session 2026-05-12 after operator's "continue" directive. C03 fix landed in 4d27c07; the same session investigated whether the residual C02 failure was Lava-fixable. CF mitigation succeeded; the remaining parser failure surfaced a separate bug that's deferred to a follow-up SP for rutacker HTML parser refresh.

2026-05-12 — c16-stale-assumption-bluff + parser-selector-fallback + firebase-test-improvement

These three changes are landed together as the anti-bluff-audit response to the operator's SIXTEENTH §6.L invocation on 2026-05-12.

Root cause 1 (Challenge16 stale-assumption bluff):

Challenge16ApiSupportedFilterTest asserted "Internet Archive must NOT appear in the onboarding provider list" because at the time of writing (Phase 12 α -hotfix) ArchiveOrgDescriptor.apiSupported was false. Phase 2b (Lava-API

per-provider routing) later flipped that to `true` without updating the test. The test continued to pass green because its `waitUntil` clause accepted the Welcome screen — where no provider list renders — making "Internet Archive absent" trivially true. Textbook §6.L bluff: green-light while the actual behavior was the OPPOSITE of the claim.

Root cause 2 (`GetCurrentProfileUseCase.parseUserId` brittle selector): The parser used a single Jsoup selector `#logged-in-username` to extract the rutracker user-id post-login. With the Cloudflare anti-bot mitigation landing in commit `f7d0a62`, the login POST now reaches a `302→200` chain successfully — but the post-login page `rutracker.org` serves to the mitigated client doesn't contain `#logged-in-username` (selector stale or mobile-shaped variant). Single-selector parsers fail catastrophically when the served HTML changes; multi-selector fallback gives graceful degradation.

Root cause 3 (`FirebaseAnalyticsTrackerTest` verify-only assertions): Two tests in `FirebaseAnalyticsTrackerTest` used `verify { mock.foo() }` as the SOLE assertion — Forbidden Test Pattern per §6.L clause 4 ("Verification-only assertions"). Call-counting is permitted as a secondary signal, never the primary one.

Affected files:

- `app/src/androidTest/kotlin/lava/app/challenges/Challenge16ApiSupportedFilterTest.kt`
- `core/tracker/rutracker/src/main/kotlin/lava/tracker/rutracker/domain/GetCurrentProfileUseCase.kt`
- `app/src/test/kotlin/digital/vasic/lava/client/firebase/FirebaseAnalyticsTrackerTest.kt`
- `CLAUDE.md`, `AGENTS.md`, `lava-api-go/CLAUDE.md` (§6.L count + 16th invocation forensic anchor)
- `docs/CONTINUATION.md` (audit findings + verification log)

Fix:

1. C16 rewritten to navigate to "Pick your providers" and assert that all 4 `verified+apiSupported` providers (RuTracker.org, RuTor.info, Internet Archive, Project Gutenberg) actually render in the list. Falsifiability rehearsal documented in the KDoc: setting `archiveorg apiSupported=false` causes the IA assertion to fail.
2. `GetCurrentProfileUseCase` now tries 4 selectors in order before erroring: `#logged-in-username`, `a.logged-in-as-username`, `.menu-userctrl a[href*='profile.php?u=']`, `a[href*='profile.php?u=']`. Error message updated to "logged-in user-id not found — page may be guest, or selectors stale" so future agents have a clearer diagnostic anchor.
3. `FirebaseAnalyticsTrackerTest`'s verify-only tests refactored to use `mockk slot captures with assertEquals` on the captured String values + `assertTrue(slot.isCaptured)` as a non-zero-count gate. The captured-value equality is now the primary assertion, per §6.L clause 4.
4. Constitutional count update in `CLAUDE.md` §6.L (THIRTEEN → SIXTEEN), `AGENTS.md` §6.L (THIRTEEN → SIXTEEN), `lava-api-go/CLAUDE.md` §6.L (TEN → SIXTEEN). Forensic anchor for the 16th invocation appended to the wall.

Verification test/challenge:

- C16 rewritten: PASS on CZ_API34_Phone API 34 live emulator (2026-05-12 10:50, BUILD SUCCESSFUL).
- 14 of 23 Challenge Tests verified PASS on live emulator this session: C00, C01, C03, C11, C12, C13, C14, C15, C16(rewritten), C20, C21, C22 (in isolation; sweep-mode fails due to C21 back-press state leak — not a real-user bug), C23, C24.
- C02 still does not pass end-to-end (parseUserId fails even with fallback selectors). The Cloudflare-mitigation portion of C02 is fully verified working.

Fix commit: (*this commit*)

Forensic anchor: operator's SIXTEENTH §6.L invocation, 2026-05-12, verbatim: "Do EVERYTHING NOW!!! ... Make sure that all existing tests and Challenges do work in anti-bluff manner ... they MUST confirm that all tested codebase really works as expected! ... execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"